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NEXT GENERATION GENETIC COUNSELING

Three Clinical Experiences with SNP Array Results Consistent with Parental Incest: A Narrative with Lessons Learned

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Abstract Single nucleotide polymorphism microarrays have the ability to reveal parental consanguinity which may or may not be known to healthcare providers. Consanguinity can have significant implications for the health of patients and for individual and family psychosocial well-being. These results often present ethical and legal dilemmas that can have important ramifications. Unexpected consanguinity can be confounding to healthcare professionals who may be unprepared to handle these results or to communicate them to families or other appropriate representatives. There are few published accounts of experiences with consanguinity and SNP arrays. In this paper we discuss three cases where molecular evidence of parental incest was identified by SNP microarray. We hope to further highlight consanguinity as a potential incidental finding, how the cases were handled by the clinical team, and what resources were found to be most helpful. This paper aims to contribute further to professional discourse on incidental findings with genomic technology and how they were addressed clinically. These experiences may provide some guidance on how others can prepare for these findings and help improve practice. As genetic and genomic testing is utilized more by non-genetics providers, we also hope to inform about the importance of engaging with geneticists and genetic counselors when addressing these findings.

Keywords SNP microarray · Microarray · Consanguinity · Incidental findings · Genomics · Qualitative · Genetic counseling · Regions of homozygosity

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Introduction

Technological developments in the last decade are accelerating our understanding of genetic contributions to disease and improving diagnosis, with the most prominent examples being chromosome microarray (CMA) technologies and next-generation sequencing (NGS). However, each development brings new ethical, legal, and psychosocial implications with an increased ability to look at larger areas of the genome in a shorter amount of time. New testing technology will continue to present challenges in the form of incidental findings (IFs) and variants of unclear significance (VUS). What constitutes an incidental finding may be unclear since many genomics tests are exploratory in nature (Cho 2008; Lohn et al. 2013a, b; McGuire and Lupski 2010). It could be argued that no finding is purely "incidental" with the broad scope of genomics tests (Cho 2008), and incidental findings are often defined variably (Downing et al. 2013). Debate continues as to what kinds of VUS and IFs should be reported to patients and research participants (Berg et al. 2011; Cassa et al. 2012; Hayden 2012), and because of this, patient education and informed consent will continue to carry crucial importance for most genetic tests.

Chromosome microarrays have brought about higher-resolution genomic analysis and improved the ability to detect dosage changes of certain and sometimes uncertain significance (Bruno et al. 2009; Miller et al. 2010; Shaffer et al. 2006). Like NGS, caution accompanies the use of CMA technology in the form of informed consent and pre-test counseling, identifying VUS, and IFs unrelated to the patient phenotype (Boone et al. 2013). Studies have looked at cancer-predisposition IFs by CMA (Adams et al. 2009; Pichert et al. 2011), and others have addressed the ability for CMAs to identify adult-onset disease. This shows that further discussion of all potential IFs found by CMA is warranted (Boone et al. 2013). These findings can have significant impact on patients and families as they struggle with unexpected findings and anxiety accompanying results (Fanos 2012). Other reviews are available that address IFs in genetic testing overall (Cho 2008; Kohane et al. 2012), including IFs in the context of CMA (Boone et al. 2013; Johnson et al. 2010). Guidelines for the use of CMA have been available for about 3 years (Miller et al. 2010); a more recent review of CMA use in the pediatrics setting is also available (Ellison et al. 2012).

In addition to identifying dosage changes, single nucleotide polymorphism microarrays (SNP-CMAs) have the added ability to detect long, uninterrupted regions of homozygosity (ROH) that can indicate uniparental disomy (UPD) or evidence of identity by descent (Grote et al. 2012; Li and Andersson 2009; Schaff et al. 2011). There are methods for calculating the percentage of homozygosity based on ROH (Papenhausen et al. 2011) and for classifying the likely degree of parental consanguinity (Sund et al. 2013). Each laboratory may differ in their methods and in their phrasing of results. It is important to note that significant amounts of ROH may provide molecular evidence of parental consanguinity, but it does not mean ROH is automatically equated with consanguinity. The detection of ROH has allowed clinicians to detect additional mechanisms of disease (like UPD) and to scan genomic regions for possible homozygous autosomal recessive mutations via autozygosity mapping of candidate genes (Alkuraya 2010; Sund et al. 2013).

However, SNP-CMAs can incidentally detect parental consanguinity (Sund et al. 2013). We have defined consanguinity as an incidental finding for SNP-CMAs since it is typically not the sole purpose for ordering these tests, and identifying consanguinity often has implications extending beyond a patient's medical issues (Downing et al. 2013). Additionally, these findings are often unexpected for those who do not realize the ability for SNP-CMA to detect parental consanguinity. Most states in the U.S. have legislation pertaining to consanguineous relationships (Harper 2010; Paul and Spencer 2008). For the Virginia Commonwealth, incestuous relationships are punishable as Class 1 misdemeanors if consensual or Class 5 felonies if aggravated based on subsection B of State Code § 18.2-366 (National District Attorneys Association 2013; Virginia General Assembly 1993).

In this paper we will utilize the term "incest" to denote relationships of first-degree consanguinity (i.e. sibling-sibling and parent–child relationships), but we do recognize the stigmatizing use of this term. In the United States, incestuous relationships are currently illegal in all states (Harper 2010), but there may be differences from state to state regarding the legality of first-cousin consanguinity. It must also be emphasized that there is cultural and religious variation in acceptance of consanguinity worldwide (Bittles 2001), and the ethical-legal issues with consanguinity may differ between cultures. Although a further discussion of consanguinity is beyond the scope of this paper, NSGC Practice Guidelines for counseling patients and couples for consanguinity are available (Bennett et al. 2002). These guidelines do not currently include information specific to incidental consanguinity found by CMAs. Additionally, incest may more commonly be the result of abuse or coercion, and there will be great concern in this context for minors and individuals with intellectual disability. However, there may be other cases of consensual incestuous relationships that are not the result of abuse or coercion, and some contemporary cases have generated debate about reproductive freedom and legal implications in Western societies (Farrelly 2008).

Some guidelines for reporting suspected consanguinity with genomic testing have been recently published (Rehder et al. 2013), but many labs may vary in how they report these findings on SNP-CMA (Grote et al. 2012; Sund et al. 2013). Additionally, there is a paucity of published literature on clinical experiences with consanguinity found incidentally by SNP-CMA (Schaff et al. 2011). A recent publication discussed the clinical and psychological implications of identifying unexpected consanguinity with SNP-CMA in one case report (Tarini et al. 2013). The authors highlight the significant ethical, legal, and psychosocial issues including discussion of personal identity, sexual abuse, reporting these findings in the medical record, misattributed paternity, the clinician's legal and moral obligations, informed consent, and pre-test counseling.

As SNP-CMA testing becomes more routine and ordered by non-genetics professionals it will be increasingly important to discuss 'consanguinity-as-incidental finding' (CIF) in terms of how clinics handle them and what resources were found to be helpful. Further awareness and discussion of CIFs can help providers ensure proper pre-test counseling and informed consent. However, extensive pre-test counseling may be overwhelming for families or impractical in certain settings (Dondorp et al. 2012; Reiff et al. 2012). Experiences with CIFs will also carry over to other genomic technologies such as NGS, since similar results are possible when using trios for whole-exome sequencing. This paper introduces three cases and qualitatively discusses clinical experiences of incestuous CIFs found by SNP-CMA, outlining some of the major ethical, social, and cursory legal issues involved. It becomes increasingly important for genetic counselors and clinicians to discuss experiences with CIFs as this dialogue can help motivate further collaborative recommendations and improve clinical practice and future research.

Methods

The information provided in this article includes discussion and qualitative analyses of three experiences ($n=3$) with parental CIFs identified on SNP-CMA. Each case involved incestuous (first-degree consanguinity) parental relationships.

Specific clinical and phenotypic information has been limited, and we also limited demographic information like gender and age to also protect privacy. However, we recognize that gender and age are both important for ethical-legal discussion. Rather than discussing clinical details, our focus will be on a narrative analysis that includes our observations, reactions, and responses surrounding the discovery of the CIFs.

Results

Introduction to the Cases

The SNP-CMAs were ordered for our patients for developmental delay or congenital anomalies. Each case was independent and involved unique medical and psychosocial factors and situations.

Case 1

The proband was an infant with prenatal findings of congenital heart defect and nonspecific dysmorphic features. The mother raised parental consanguinity as a possibility when she met with a prenatal genetic counselor. This was unconfirmed and communicated outside of the medical records, and she suggested that there may be another possible father. Routine karyotype and metabolic testing were normal. A SNP-CMA showed no copy-number changes, but it identified several ROH (each > 5 Mb in size) with total genomic homozygosity estimated to be 601.5 Mb. According to the CMA report, this was most consistent with parental relatedness. A family history was obtained with a pediatric genetic counselor prior to ordering the SNP-CMA, and consanguinity was denied at that time. The mother was a legal adult at the time of the patient's conception. It was only until after the results of the CMA were available that the mother revealed a consensual relationship with her father. Other family members were not aware of this relationship.

Case 2

An older pediatric patient was referred for a possible genodermatosis. The patient also had developmental delay but no striking dysmorphic features. Multiple family members were reported to have learning disabilities, psychiatric disorders, and possible substance abuse. There was no information available about the proband's father, and the mother denied consanguinity. To further investigate the proband's developmental delay, a SNP-CMA was ordered and found no copy-number changes. It identified several ROH (each > 8 Mb in size) with total genomic homozygosity estimated to be 687 Mb. According to the CMA report this was most likely due to first-degree parental consanguinity. The mother was an adult at the time of the patient's conception. During post-test counseling she revealed a previous father-daughter incestuous relationship.

Case 3

The proband was a baby referred for multiple issues including neuromuscular problems, congenital heart defect, abnormal EEG, and minor dysmorphic features. Radiologic studies were normal, as was a karyotype. Initial family history was unremarkable, and no information about the father was available. A SNP-CMA revealed normal copy-number but identified multiple ROH (each > 8 Mb) with the total homozygosity size estimated to be 775 Mb. The CMA report described this as most consistent with parental first-degree consanguinity. The infant continued to have significant respiratory distress and died. Family history showed a pattern of learning disabilities, and the baby's mother may have had a mild learning disability. Consanguinity was denied at this time. It was later found that another sibling had a separate evaluation at a different medical institution, and a SNP-CMA for this sibling also revealed multiple ROH totaling approximately 600 Mb. This information was not known at the time of our evaluation of the proband. The mother was a young adult at the time of the proband's conception.

Discussion

Clinical Responses to the CIFs—Post-test Counseling

Case 1

As a consanguineous relationship was suspected based on prenatal history prior to receiving the SNP-CMA results, this was not as surprising. Since consanguinity had been denied when recording the family history, identifying consanguinity was not the purpose for doing the test, and the result has implications beyond the patient's health, we consider this to be an incidental finding. At post-test counseling we discussed that the result confirmed the report of possible consanguinity. We preferred to not use the term "incest" when counseling the family given its stigmatization, and instead, explained the meaning and significance of ROH with an emphasis on the possible increased risks for intellectual disability and autosomal recessive diseases. We stressed to the family that we were only interested in determining a diagnosis and helping to guide the child's care. However, it was unclear how the multispecialty hospital team would use this information.

Social workers and Child Protective Services had been notified by the hospital, and the findings had been reported to the facility's legal services. The clinician's role in reporting to legal services was initially unclear. We later learned that the hospital had an obligation to document and report the finding to authorities for two reasons: 1) there was a realistic concern for sexual abuse, coercion, and the involvement of other children; and 2) since incest is illegal in the Virginia Commonwealth the hospital had to comply with reporting this incident. The decisions and obligations to report the CIFs in this case were in accordance with NSGC Code of Ethics Section II part 6: “Maintain information received from clients as confidential, unless released by the client or disclosure is required by law” (National Society of Genetic Counselors 2006). The review by social services found that there were no overt signs of psychological or sexual coercion. Additionally, there was no evidence to indicate harm to other children or family members, but social workers would periodically visit the family and assess individuals for any signs of distress or abuse. At the time of writing, no formal charges have been brought.

Case 2

Consanguinity had been denied at family history, and the SNP-CMA was ordered to evaluate for developmental delay and not for the referral reason of genodermatosis (typically single-gene mutations). Although consanguinity was unexpected, we counseled on the meaning and significance of multiple ROH. The genetic counselor and physician discussed the likelihood of parental consanguinity and the increased risks for recessive diseases and intellectual disability. In terms of the developmental delay, it was discussed that the degree of relatedness was probably contributory.

During post-test counseling the mother voluntarily revealed a prior incestuous father-daughter relationship at the time of the patient’s conception. The several years between the end of the relationship and our evaluation clouded whether legal involvement was indicated. To further complicate the issue, the patient’s father was not involved at all, and contact had been lost with him years ago, thus obscuring the ability to assess for psychological or sexual abuse. The genetics team decided to not pursue the finding from a legal standpoint. The obligation to report the findings to a legal representative was vague. We provided support and information for additional resources and counseling as needed.

Case 3

Once the SNP-CMA results were available, social workers were immediately informed given the concern for abuse of the mother and the children. Social work was helpful with probing more into the social situation and acquiring the SNP-CMA results on the sibling after additional investigation into the family situation. For the other sibling’s CMA findings, the mother thought that the results were normal. It is unclear whether the mother was adequately counseled on the significance of the ROH at the previous institution or whether her comprehension of the discussion was limited. The possibility of psychological and sexual abuse by a family member was suspected and social workers and Child Protective Services were both more involved. The mother was reluctant to identify the father. The mother’s only first-degree male relative was her father, and an investigation was initiated on the suspicion that the mother had been abused on multiple occasions. The social workers had been working closely with local law enforcement given the substantiated concern for psychological, physical, and sexual abuse. At the time of writing, charges against the proband’s father have been made.

What About Pre-test Counseling?

In Cases 1 and 2, there was a discussion regarding the possible results of CMA that included copy-number changes of known and unknown significance as well as finding IFs unrelated to the patient’s phenotype (like cancer predisposition). Pre-test counseling for consanguinity was done in Case 1 due to the foreknowledge of possible parental incest. For Case 2, a genetic counselor was available before ordering the CMA for this discussion. Since consanguinity was denied in Case 2 and there was no prior suspicion, pre-test counseling did not cover CIFs. For Case 3, the CMA was ordered without appropriate pre-test counseling.

Our Reactions: Pre- and Post-test Counseling, Psychosocial Factors, Social/Legal Factors

For Case 1, the pre-test discussion included a brief review of consanguinity, and this was stressed because of the prior information that parental incest was a possibility. We considered this finding incidental for three main reasons: 1) Consanguinity had been denied when we recorded the family history, 2) the purpose of ordering the CMA was not to confirm consanguinity but to identify pathogenic copy-number changes, and 3) the results would have wider legal and psychosocial impacts beyond the medical issues of the proband. We discussed the ROH at post-test counseling, and the medical record still documented the large amount of ROH. The mother and her father had requested that their relationship not be documented. The main question was why the family went forward with the SNP-CMA knowing the possible test results. Perhaps the inpatient hospital setting is not an ideal place to safely and privately discuss consanguinity, and perhaps the family would have made a different decision at an outpatient visit. Further discussion is warranted for how to best discuss consanguinity in an inpatient setting. Since we were alerted to possible parental incest by the prenatal genetic counselor beforehand, our team was much more prepared to deal with this CIF. However, the prenatal counselor’s suspicion of possible consanguinity was in conflict with our family history of denied consanguinity, and this discrepancy raises several counseling questions pertaining to trust and confidentiality.

For similar cases, a discussion of the benefits and harm of reporting results like these to legal representatives is warranted. A case could be made on grounds of beneficence and non-maleficence of not reporting this finding to authorities in cases of consensual consanguinity. Other cases could be made that not reporting it may also cause harm pertaining to abuse. It is important to fully discuss the significance of these findings with the parents. We do not agree with limiting full disclosure of CIFs, and this opinion is in accordance with the NSGC Code of Ethics for responsibilities to clients. Relevant information in the Code of Ethics Section II part 4 states that genetic counselors should "Enable their clients to make informed decisions, free of coercion, by providing or illuminating the necessary facts, and clarifying the alternatives and anticipated consequences" (National Society of Genetic Counselors 2006). The Code of Ethics in Section II part 3 also states that genetic counselors should "Respect their clients' beliefs, inclinations, circumstances, feelings, family relationships and cultural traditions" (National Society of Genetic Counselors 2006), and this also has relevance for cases of CIFs and the decision to report these findings to authorities in cases of both consensual and non-consensual consanguinity.

For Case 2, pre-test counseling for SNP-CMA did include IFs, but did not include a discussion of revealing parental consanguinity. In hindsight, this was based on the mother denying parental consanguinity. As in Case 1, we addressed the meaning and significance of ROH without the use of the term "incest". Again, our fear was that we would alienate the patient and mother by sounding accusatory. On this note, we discussed the increased risks for autosomal recessive diseases and intellectual disabilities, and we had to discuss that the CMA did not identify the cause for the original referral indication of genodermatosis. It was interesting that consanguinity was initially denied, but only after the results did the mother reveal a past possibly-consensual adult consanguineous relationship with her father. Because the patient's date of conception was several years before our evaluation, our legal obligations to report the finding were unclear. We ultimately decided to not report this finding to institutional legal services. In addition, we feared that involving legal authorities could actually bring unintended psychosocial harm to this single-parent family. Although we cannot confirm familial abuse, we remained concerned about the mother's psychosocial well-being and provided resources for support.

For Case 3, the pre-test counseling was limited due to the inpatient environment which may not have been conducive to extended discussion on CIFs. Additionally, the single mother's learning disability could have inhibited an in-depth discussion. The mother had capacity for some decision-making, but her comprehension of information may have been limited. This result was unexpected, and our preparedness for discussing and acting upon the results was less than the other cases. The involvement of social workers proved to be pivotal.

The post-test counseling occurred as a care conference in the presence of the attending hospital providers, a genetic counselor, and social workers with whom the mother had developed a close relationship. The consanguinity was communicated to the mother, although the team stressed their role in using the information to determine a diagnosis with the goal of helping her baby. The mother continued to deny consanguinity and refused to identify the father. Social workers later commented that the results of the SNP-CMA may have brought on feelings of guilt and shame, further highlighting how CIF results may alienate individuals. During the care conference the team tried to address and alleviate guilt. Eventually, the mother felt comfortable with the social workers to the point that she admitted the previous sexual abuse by one family member.

The strongest suspicion for sexual and psychosocial abuse occurred in Case 3 given the mother's mild intellectual disability, the social situation, and having at least two children with high-degree ROH. We note that the mother was a legal adult at conception with her third child, but another child with ROH would have been conceived when the mother was a minor. The social workers had better access to appropriate resources and were more able to address these from a social and legal standpoint. With the help of local authorities the social workers were able to facilitate an investigation of the family and social environment, which found that the situation was dangerous for the mother and any children involved.

These three cases serve as clear examples of the possibility of revealing CIF by SNP-CMA even in the context of denied consanguinity. Each presents a different situation and unique factors making the clinical and counseling approach to addressing the CIF more difficult. We were able to analyze and discuss several factors relevant in each case: our pre- and post-test counseling and reactions to the CIF, the psychosocial factors in each case, the legal aspects and further proceedings, and the ethical implications of the results. These cases also demonstrate the necessity of social work involvement, especially for cases of suspected psychological and sexual abuse. Interdisciplinary teams involving social workers are important for addressing these findings.

Practice Implications, Recommendations, and Lessons Learned

The cases here provided tremendous perspective on how to address unexpected CIFs by CMA. Although some of these points are mostly self-evident for our practice, it is always beneficial to readdress them as part of one's clinical routine. One of the key points to consider is that pre-test counseling about consanguinity may be limited for CMAs when parents deny consanguinity or when there is no suspicion of it in the family history. We would recommend that for CMAs pre-test counseling should always include a discussion of consanguinity. There are seven main lessons we learned from these experiences:

1) Denial of consanguinity does not mean consanguinity is improbable.
2) For families who may want sensitive information like consanguinity left out of the medical record, it is highly recommended to consult with institutional legal and privacy personnel before making any decisions about charting.
3) One should attempt to gather an overview of the social situation during the family history, delving into parent's background and education. Additional information can come from other providers or social workers.
4) It is important to discuss CIFs during pre-test counseling despite denial of consanguinity.
5) It may be important to discuss with other ordering providers about how to address potential consanguinity or IFs identified by SNP-CMA.
6) It is important that the medical genetics community continue to work on guidelines for reporting ROH and addressing consanguinity as an incidental finding in the clinical setting (Grote et al. 2012; Rehder et al. 2013; Sund et al. 2013). This includes better interaction between testing laboratories and clinical professionals.
7) These and similar cases involving CIFs should help motivate an update of NSGC Practice Guidelines pertaining to consanguinity, CIFs, and IFs in a more general context.

Additional Questions and Future Research

Throughout these experiences, several questions were raised that may be addressed by further research or interaction among the medical genetics community. In these cases, if pre-test counseling had included better discussions of CIF, would the families still choose or not choose to do the CMA? Interestingly, in a recent case report a mother said she would have considered declining a CMA for her child had she known prior that it could identify parental consanguinity (Tarini et al. 2013). Do genetic counselors have a duty to confidentiality and a duty to protect otherwise consensually incestuous couples in disagreement with state law? What if consensual incestuous couples want information omitted from the record, and how might this impact the future care of their children? This also became a key issue in the report by Tarini et al. (2013). Again, genetic counselors in similar situations should carefully weigh the options using the NSGC Code of Ethics, specifically Section II and consult with legal representatives.

Are there legal cases for individuals whose results showed unwanted report of consanguinity against their wishes? A recent survey found that there have been few legal cases regarding IFs from genomic testing, likely because of the novelty genomic testing (Clayton et al. 2013).

Conclusion

These three narratives discuss consanguinity found by SNP-microarray, and how these unexpected results can have far-reaching implications for health and development, psychosocial well-being, and for possible involvement of legal authorities. These results can have significant impact on family structure and well-being, and it is crucial that genetic counselors, geneticists, and non-genetics providers who order CMAs continue to discuss these types of results. There is some consistency among genetics professionals and the general public that IFs should be reported to patients if the results indicate a serious condition that was actionable (Berg et al. 2011; Haga et al. 2011). However, there is less agreement for IFs with social implications, like incest or consanguinity (Lohn et al. 2013a, b). As genomic technology in general is increasingly used in various clinical scenarios, it will be important to address adequate pre-test counseling and incidental findings with patients and families.

This discussion adds to the current literature on reporting and managing incidental findings. These cases also serve as excellent teaching opportunities for genetic counseling students, medical students, and for larger discussions involving bioethics, incidental findings, and genomic technology. We would also recommend that the NSGC Practice Guidelines on Consanguinity be updated to include CIFs identified by new genomic testing. Improved preparedness will hopefully diminish the distress that can accompany these results and allow providers to utilize resources for handling them appropriately.

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